Advanced Data Structures
CS4580

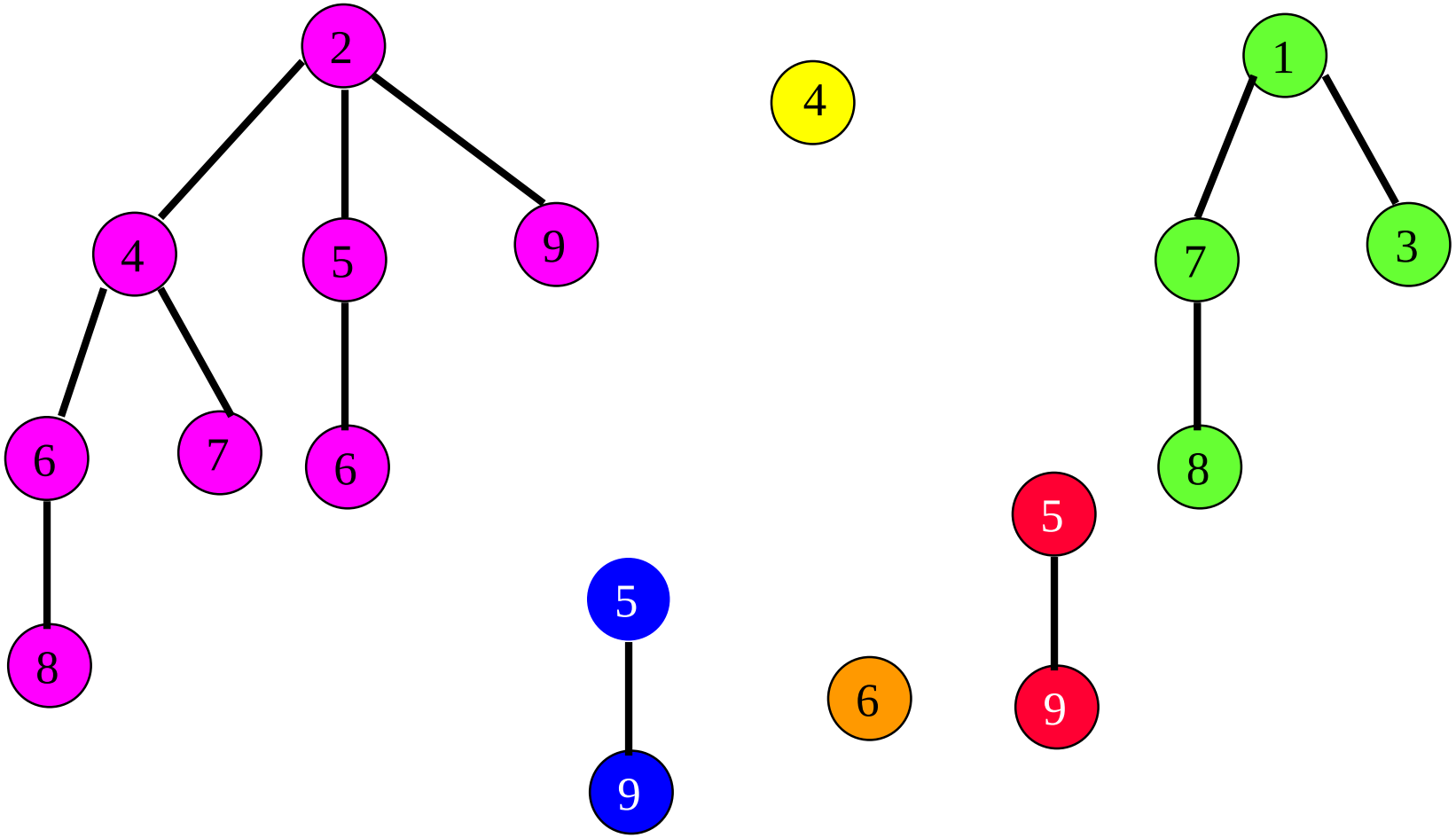
Lecture 06

Binomial Heaps

	Leftist trees	Binomial heaps	
		Actual	Amortized
Insert	O(log n)	O(1)	O(1)
Remove min (or max)	O(log n)	O(n)	O(log n)
Meld	O(log n)	O(1)	O(1)

Min Binomial Heap

- Collection of min trees.

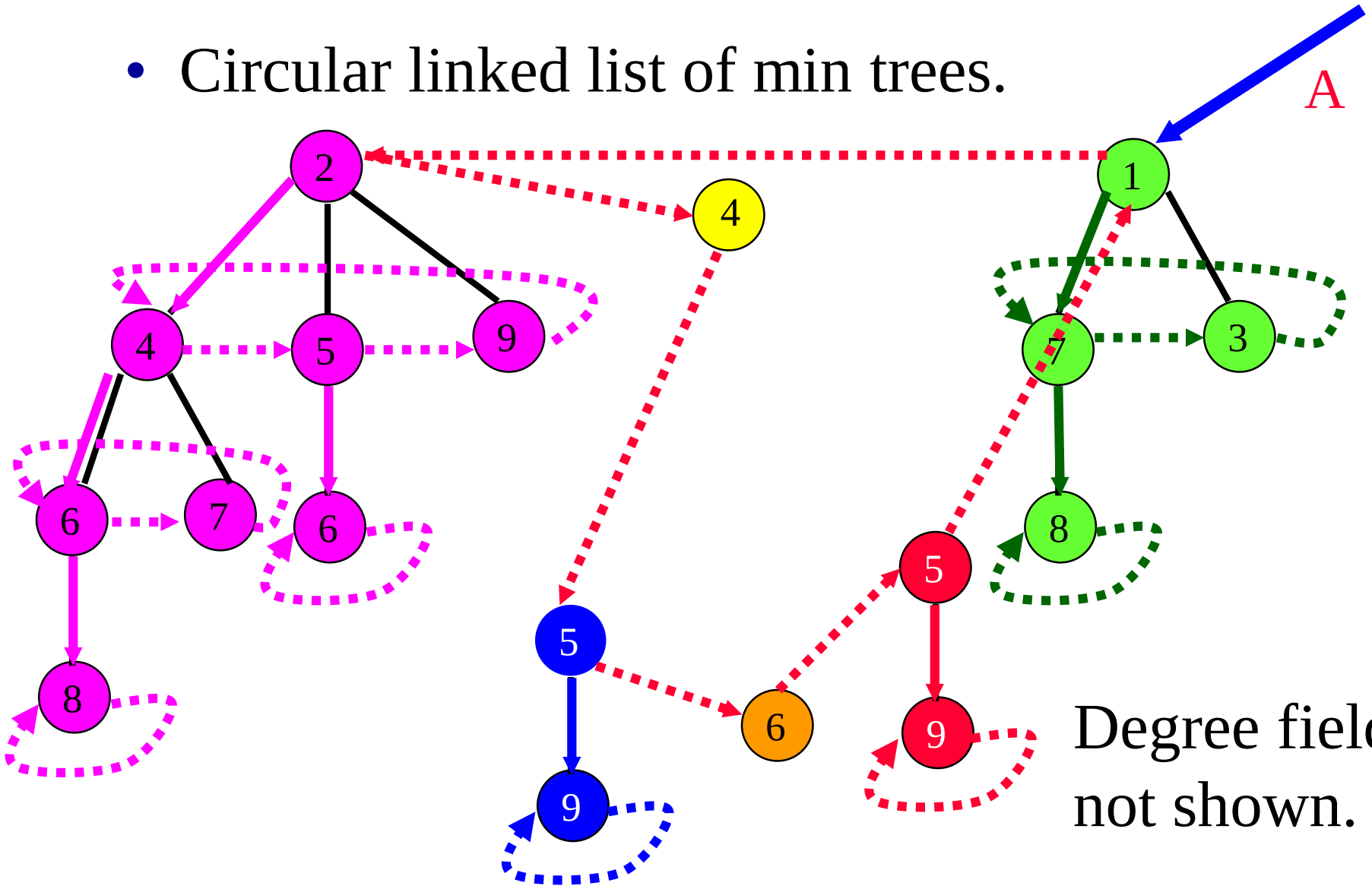


Node Structure

- Degree
  - Number of children.
- Child
  - Pointer to one of the node's children.
  - Null iff node has no child.
- Sibling
  - Used for circular linked list of siblings.
- Data

Binomial Heap Representation

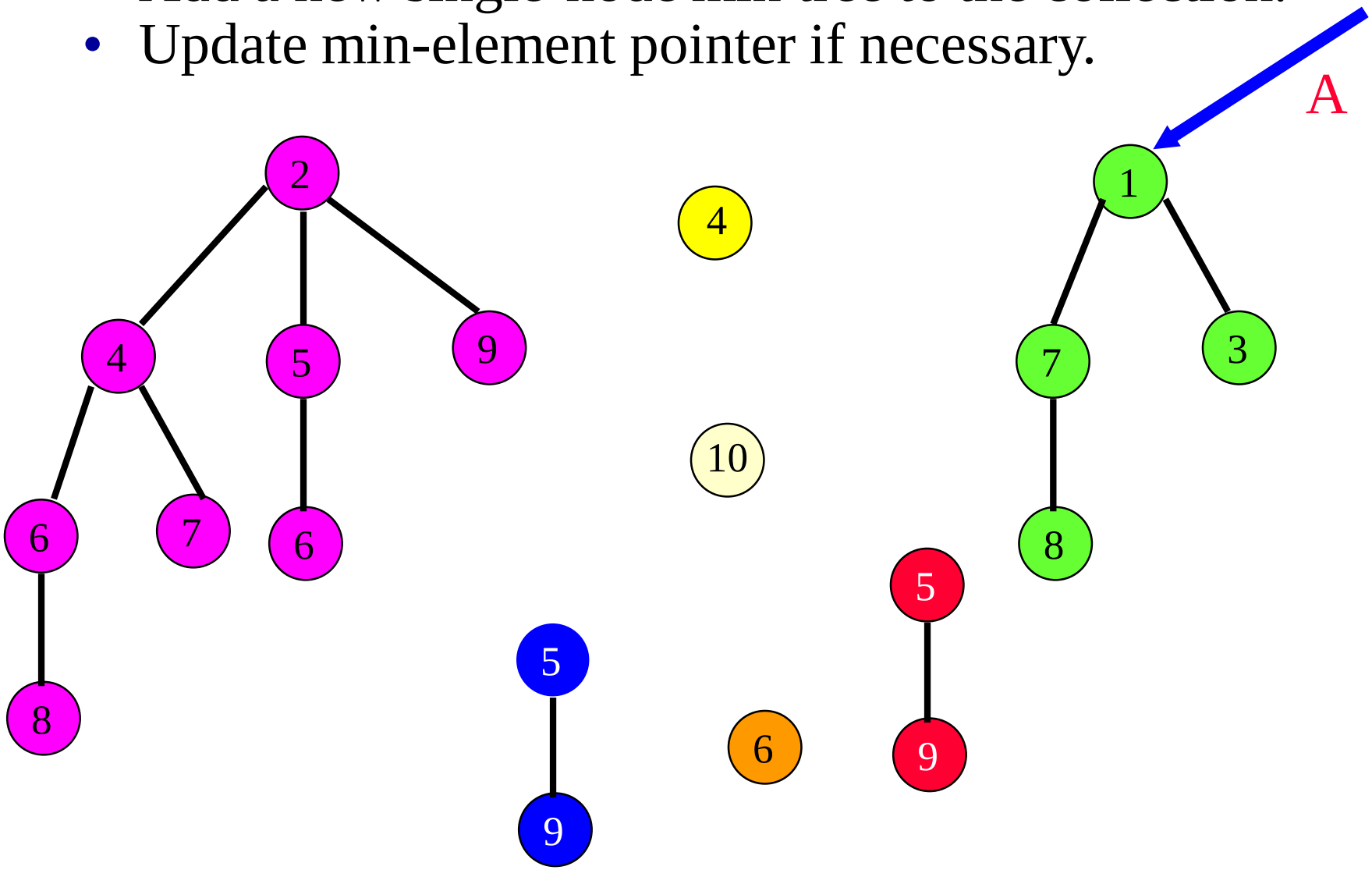
- Circular linked list of min trees.



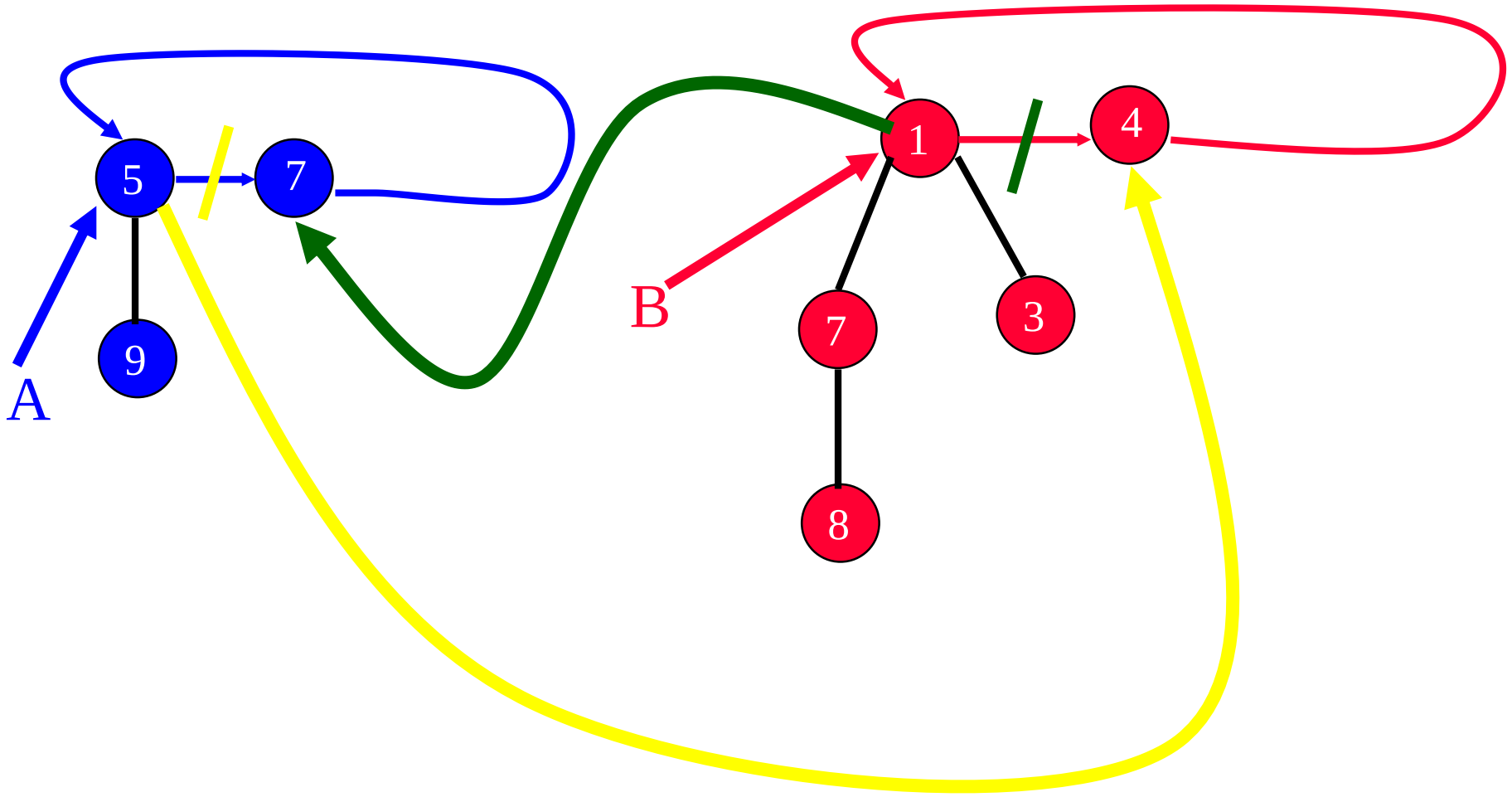
Degree fields not shown.

Insert 10

- Add a new single-node min tree to the collection.
- Update min-element pointer if necessary.

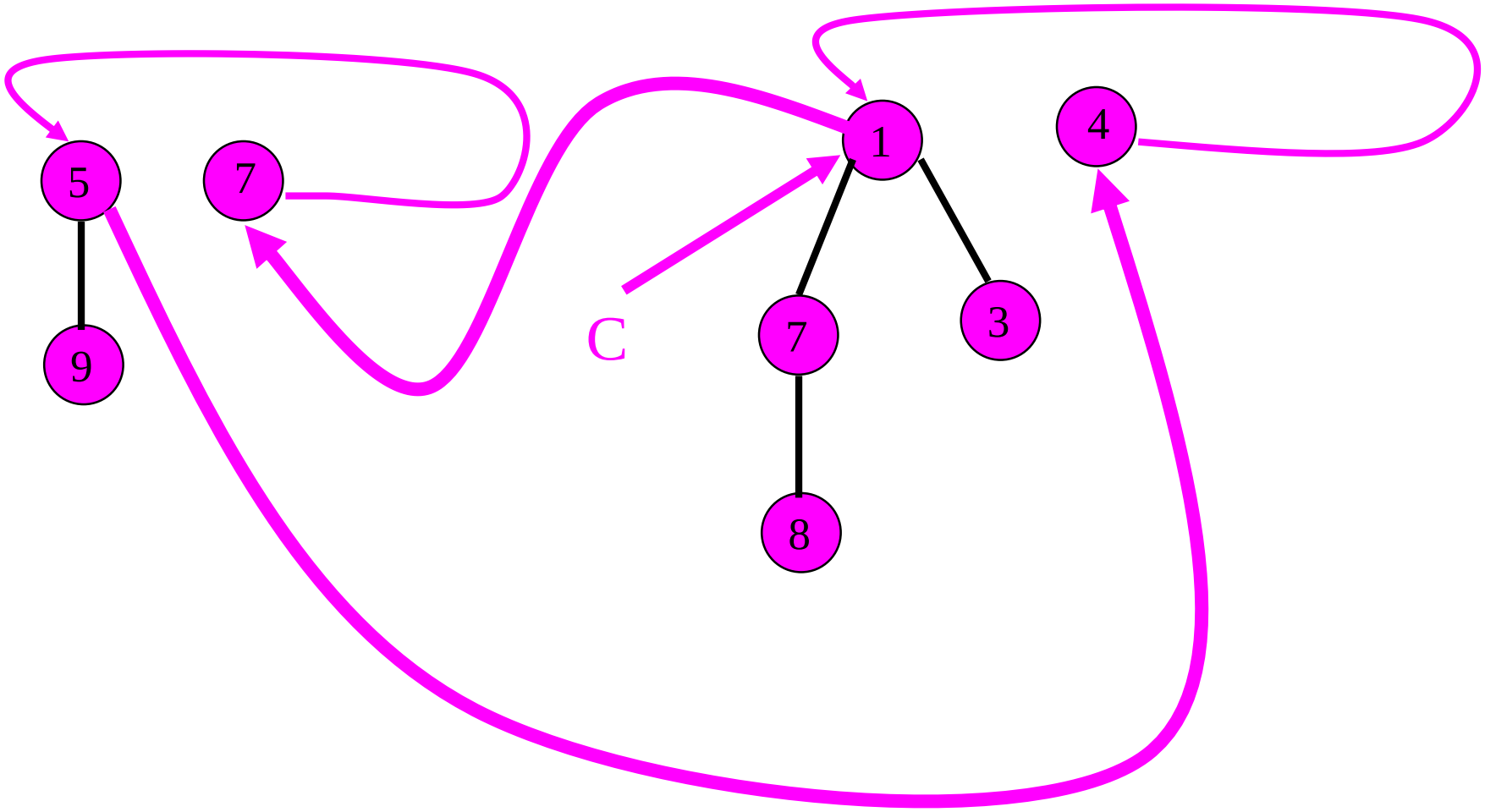


Meld



- Combine the 2 top-level circular lists.
- Set min-element pointer.

Meld

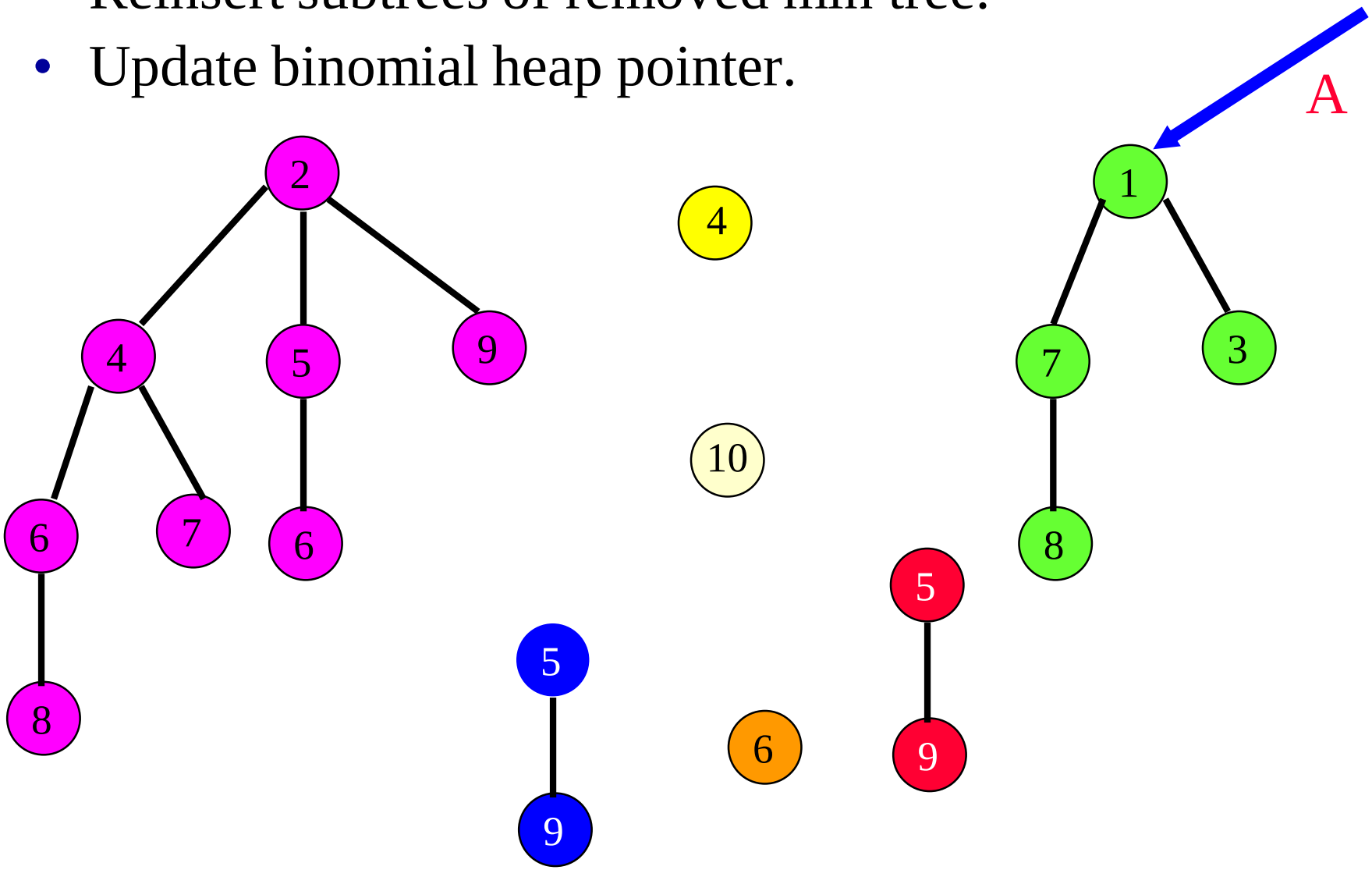


Remove Min

- Empty binomial heap => fail.

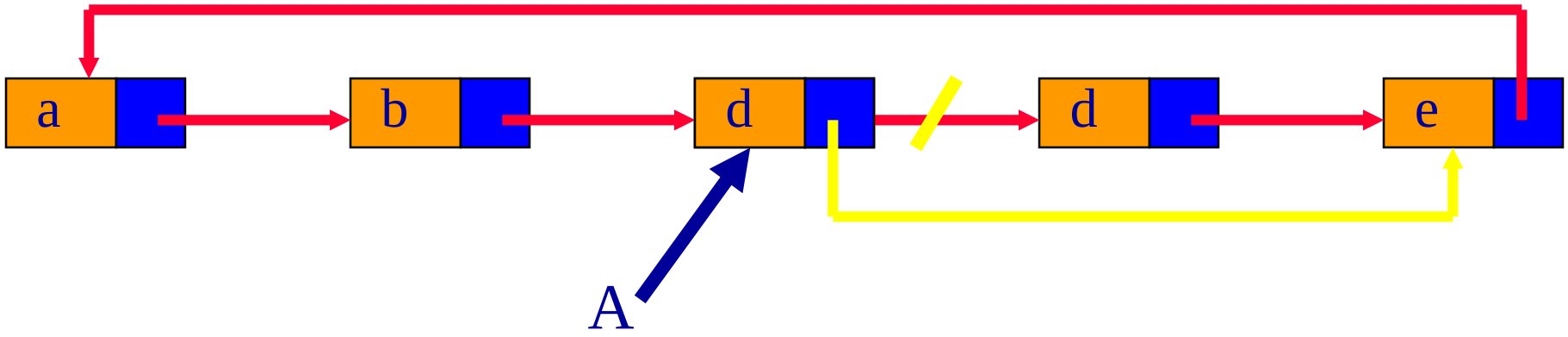
Nonempty Binomial Heap

- Remove a min tree.
- Reinsert subtrees of removed min tree.
- Update binomial heap pointer.



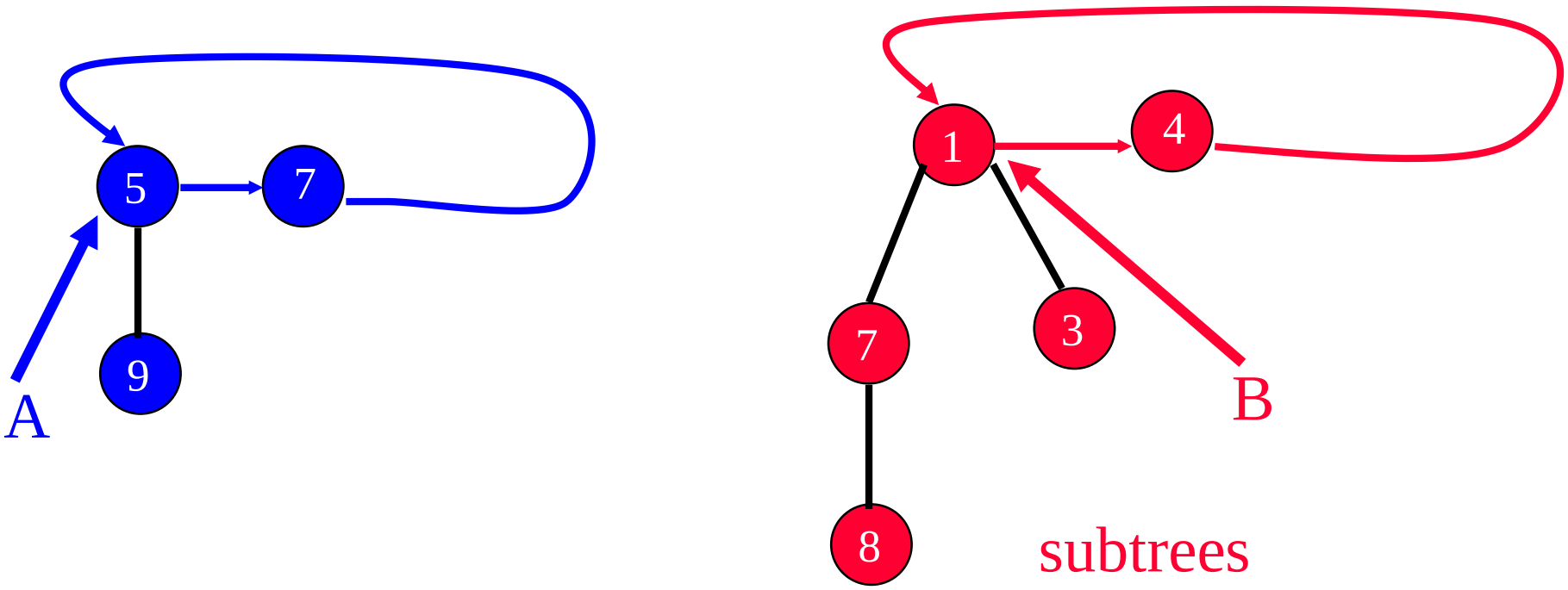
Remove Min Tree

- Same as remove a node from a circular list.



- No next node => empty after remove.
- Otherwise, copy next-node data and remove next node.

Reinsert Subtrees



- Combine the 2 top-level circular lists.
  - Same as in meld operation.

Update Binomial Heap Pointer

- Must examine roots of all min trees to determine the min value.

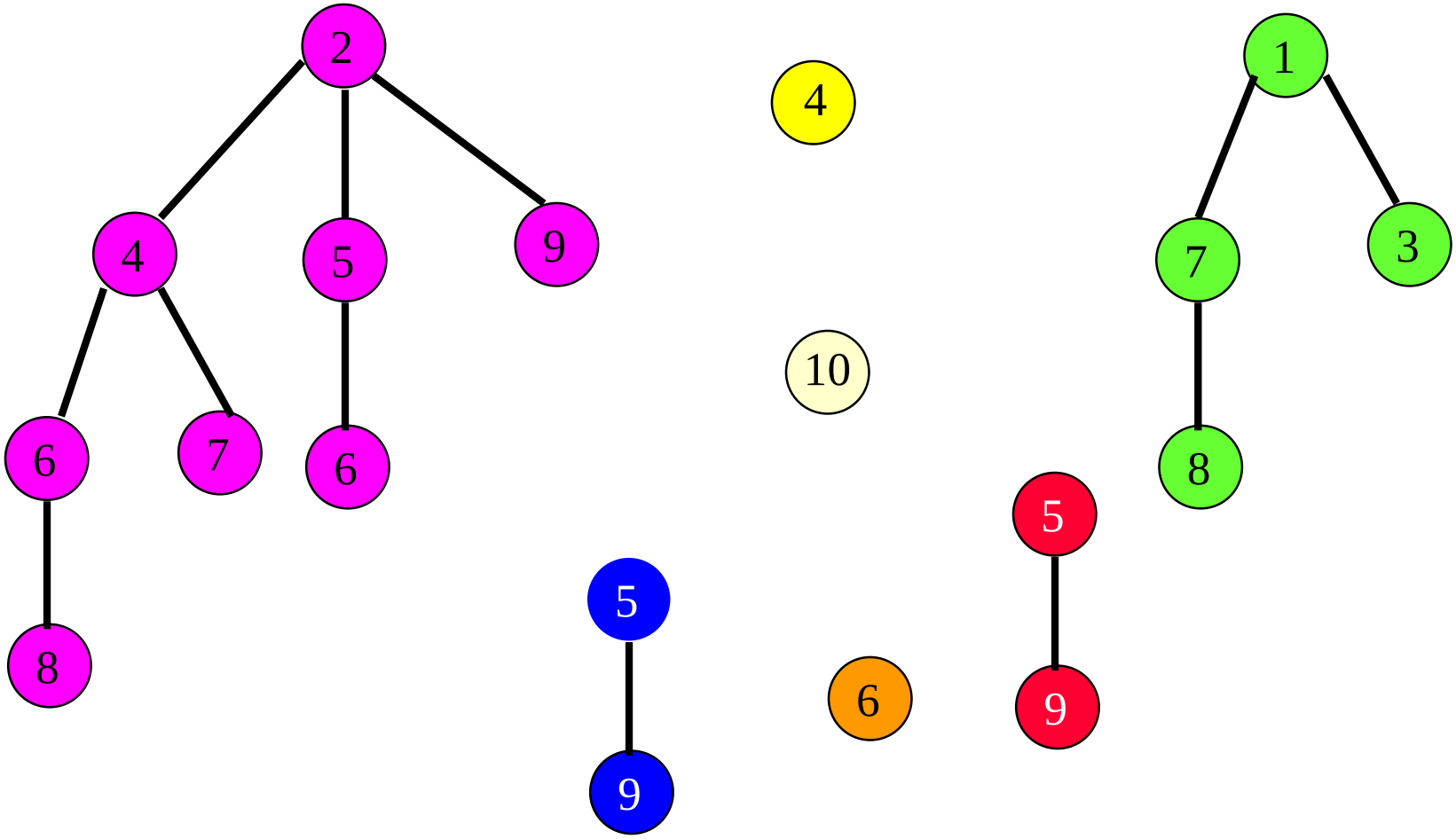
Complexity Of Remove Min

- Remove a min tree.
  - O(1).
- Reinsert subtrees.
  - O(1).
- Update binomial heap pointer.
  - O(s), where s is the number of min trees in final top-level circular list.
  - s = O(n).
- Overall complexity of remove min is O(n).

Enhanced Remove Min

- During reinsert of subtrees, pairwise combine min trees whose roots have equal degree.

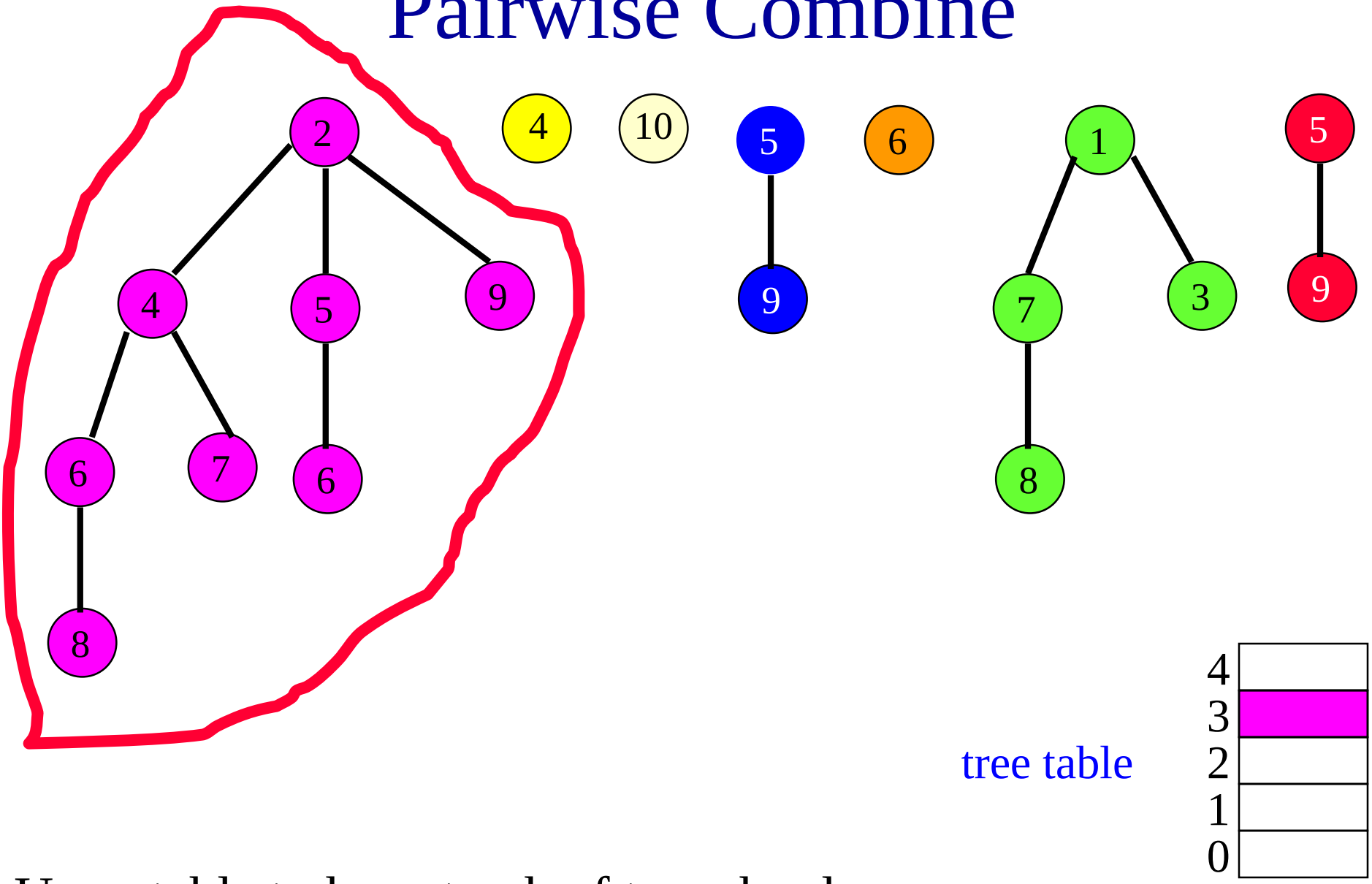
Pairwise Combine



Examine the s = 7 trees in some order.

Determined by the 2 top-level circular lists.

Pairwise Combine

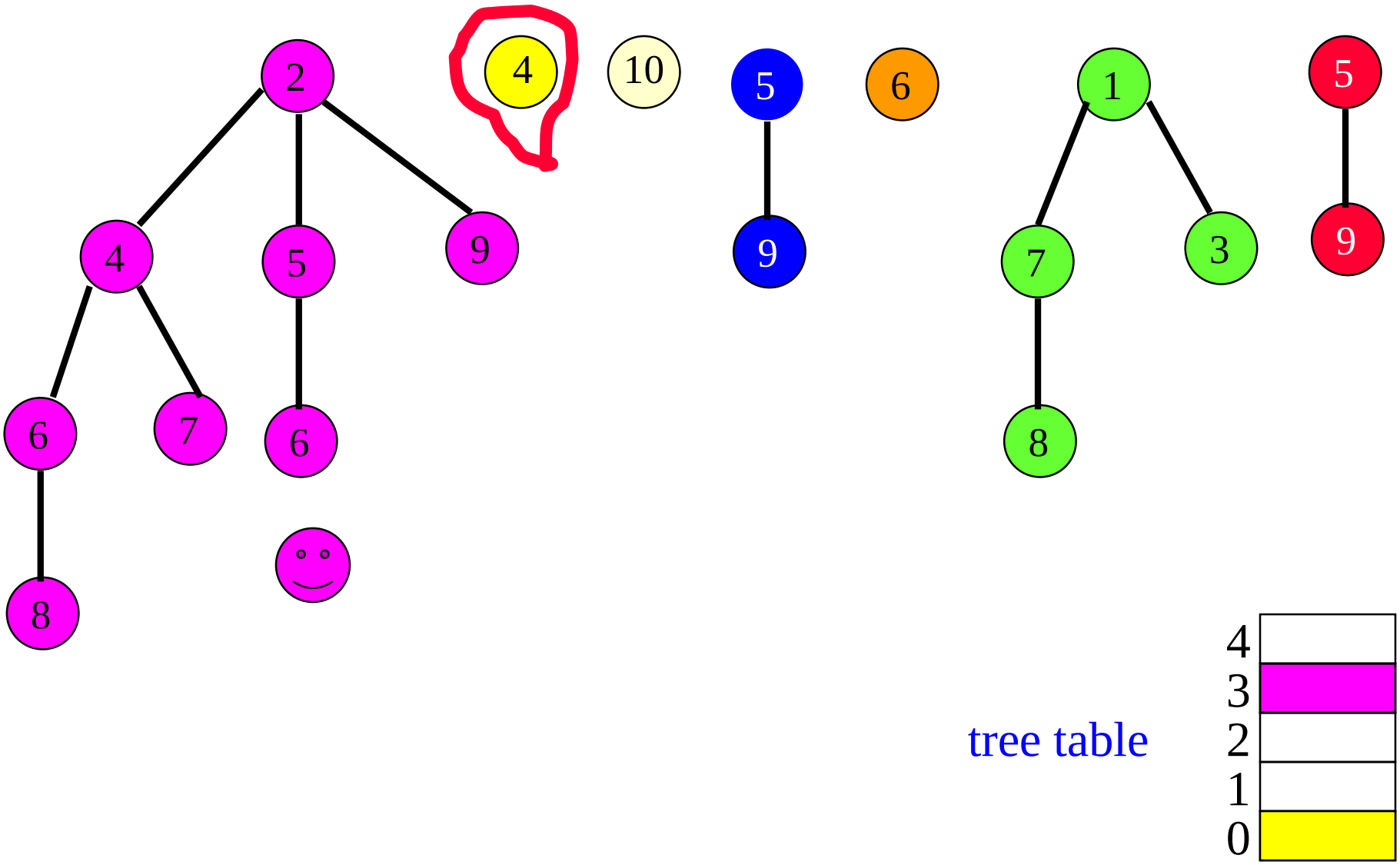


tree table

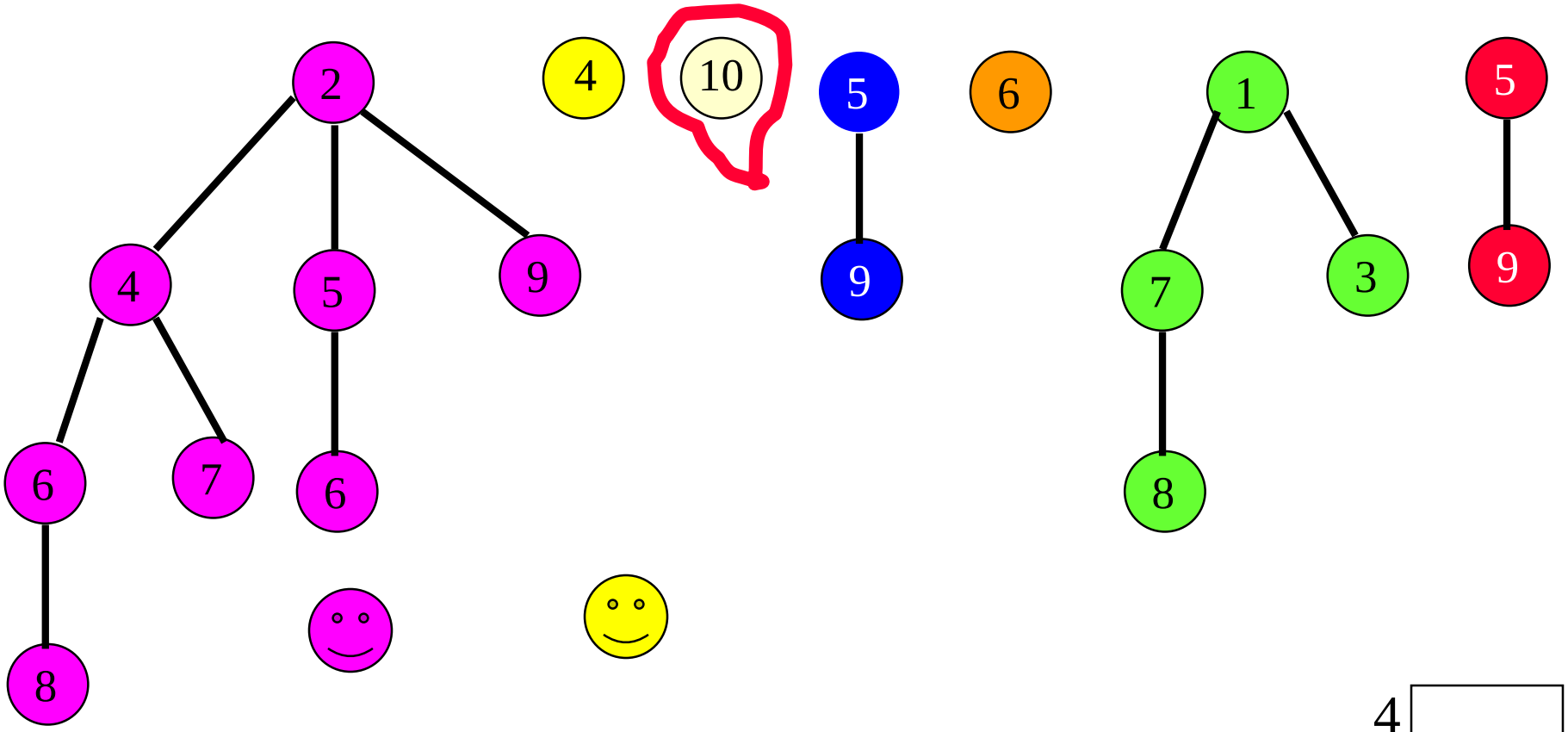
4	
3	
2	
1	
0	

Use a table to keep track of trees by degree.

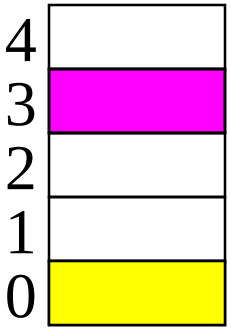
Pairwise Combine



Pairwise Combine



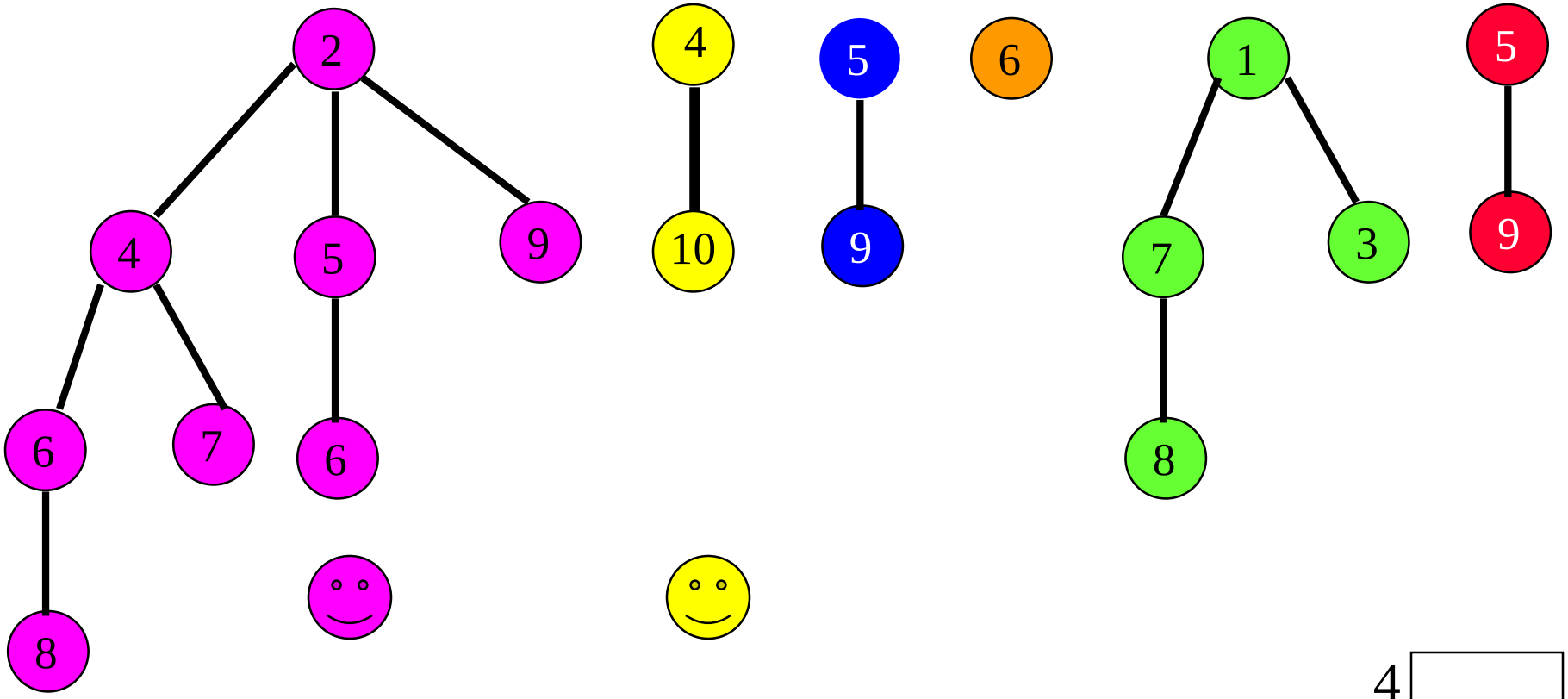
tree table



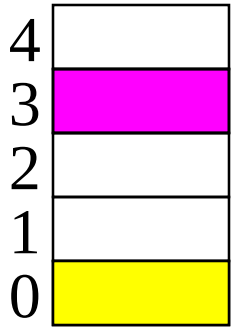
Combine 2 min trees of degree 0.

Make the one with larger root a subtree of other.

Pairwise Combine

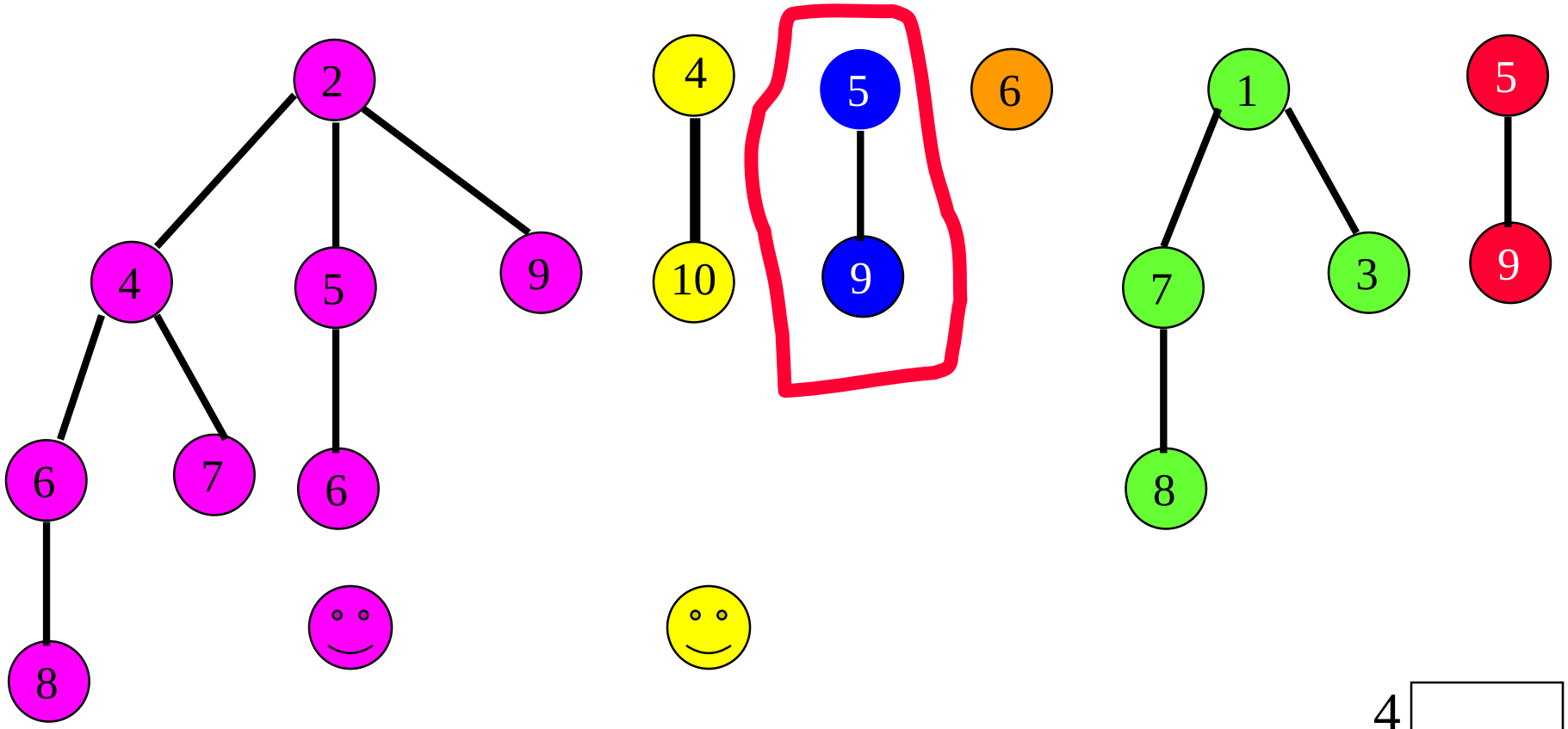


tree table

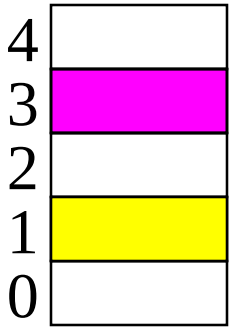


Update tree table.

Pairwise Combine



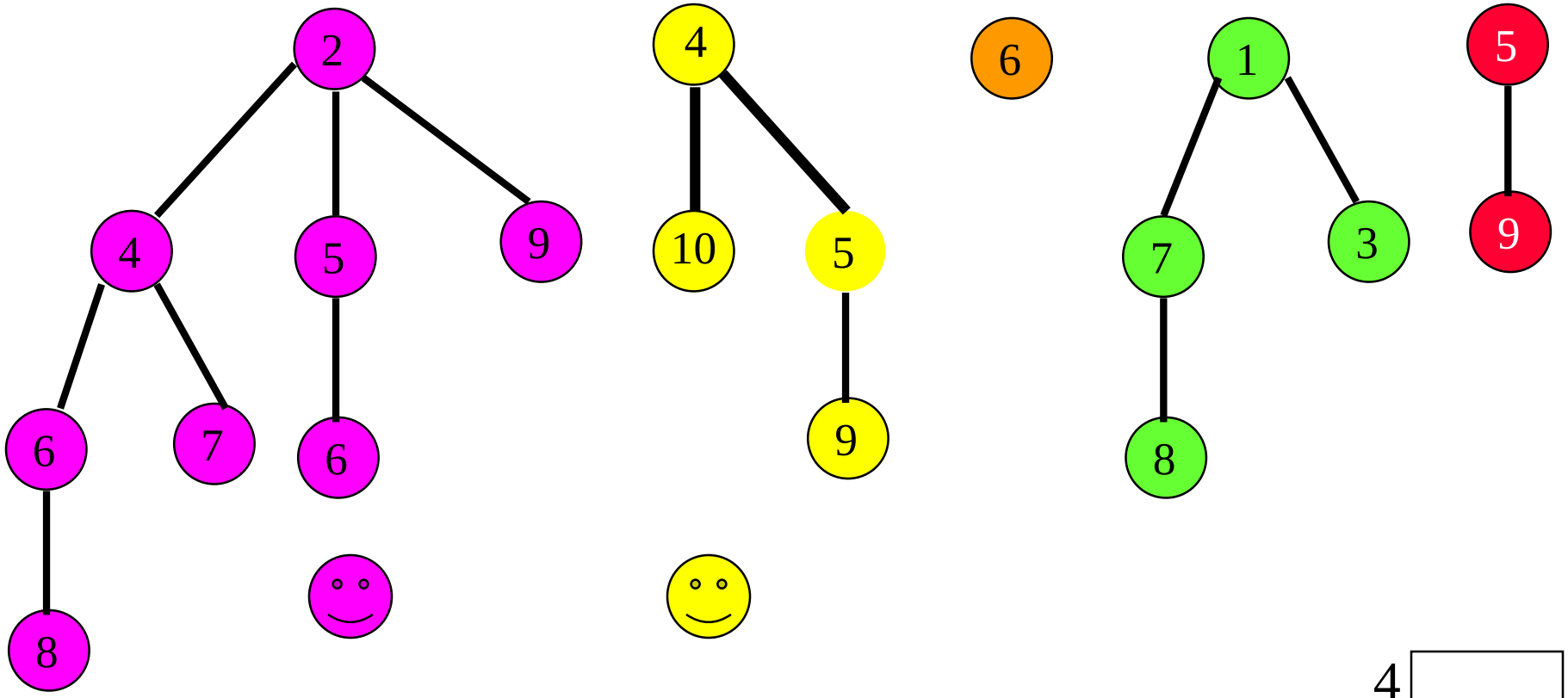
tree table



Combine 2 min trees of degree 1.

Make the one with larger root a subtree of other.

Pairwise Combine

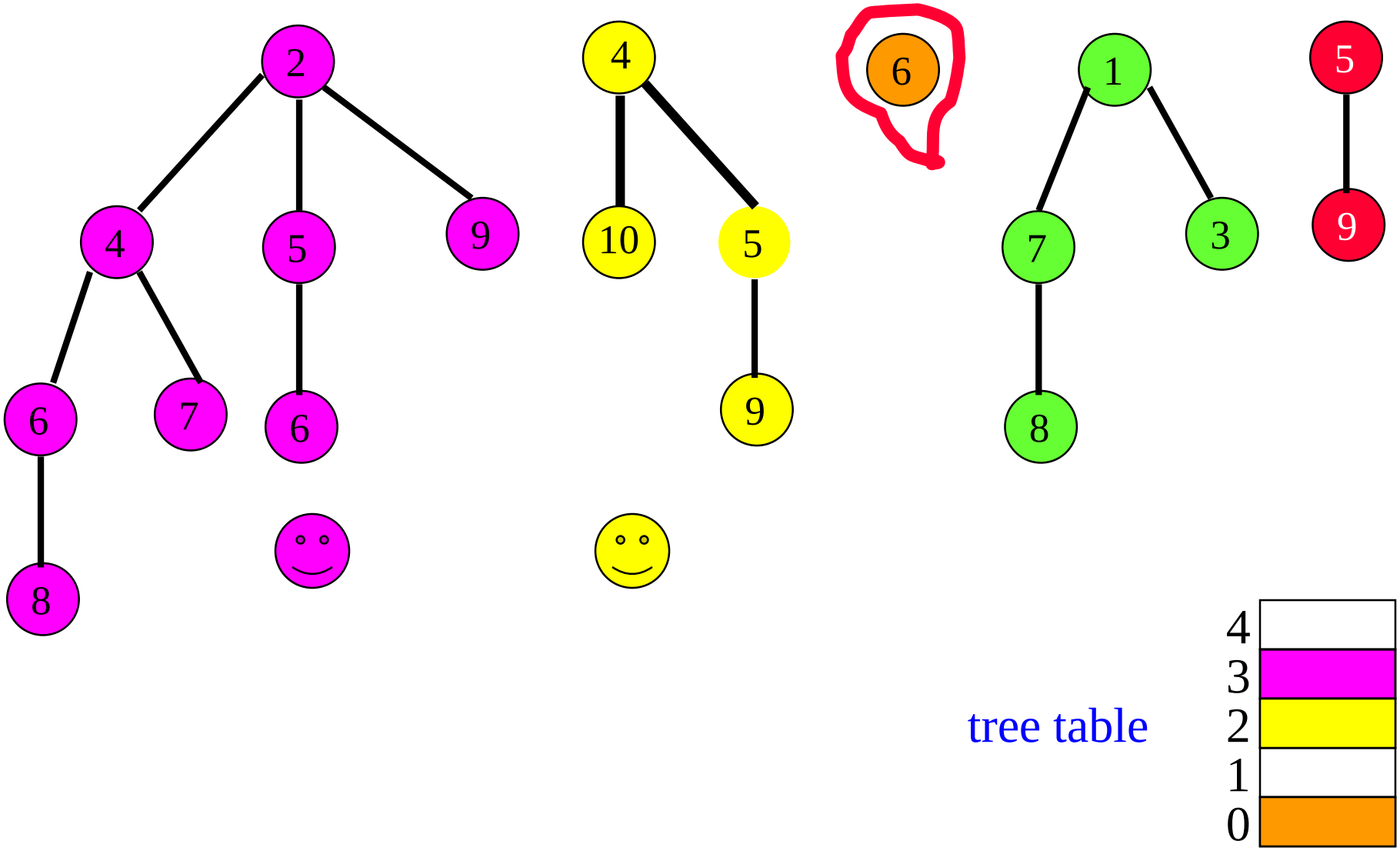


tree table

4	
3	
2	
1	
0	

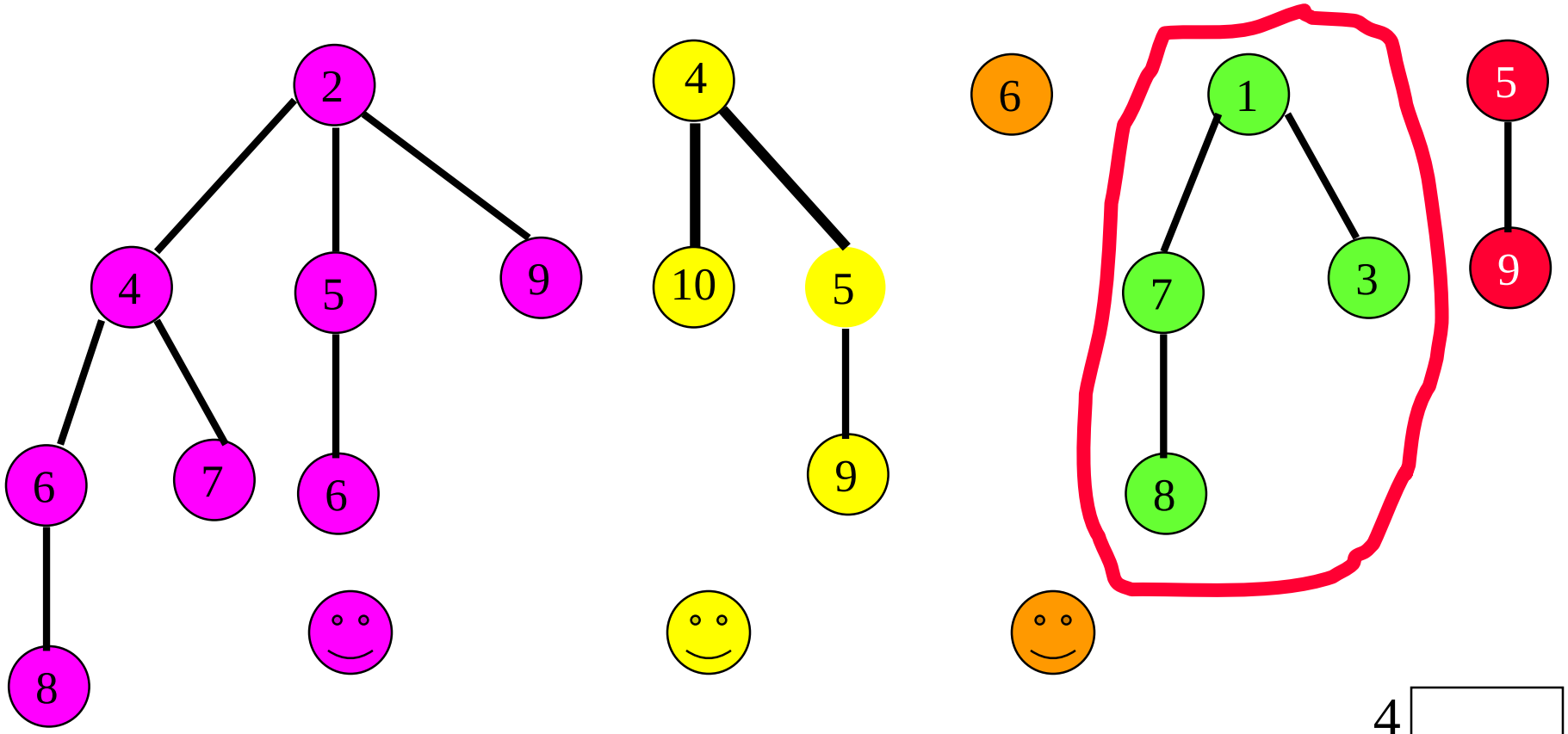
Update tree table.

Pairwise Combine

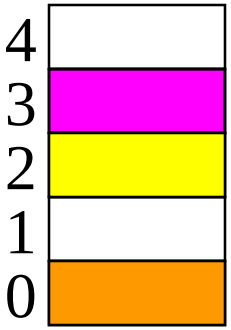


tree table

Pairwise Combine



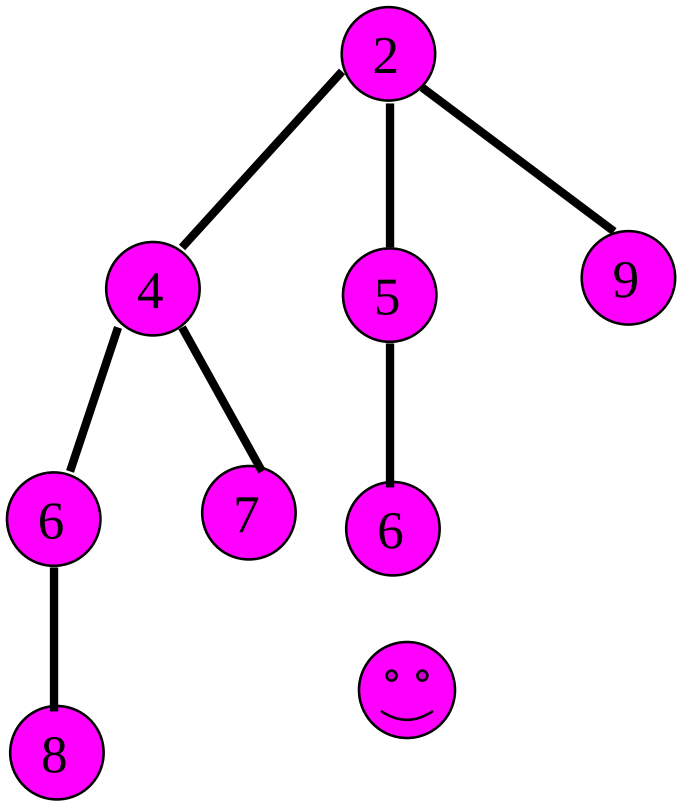
tree table



Combine 2 min trees of degree 2.

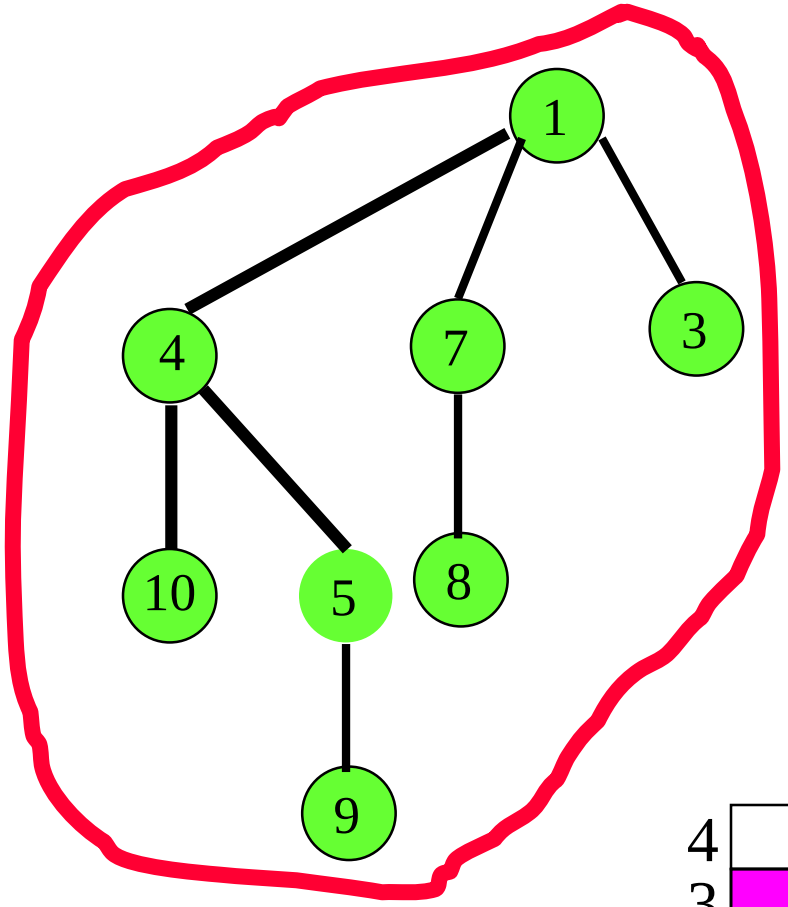
Make the one with larger root a subtree of other.

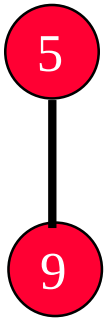
Pairwise Combine



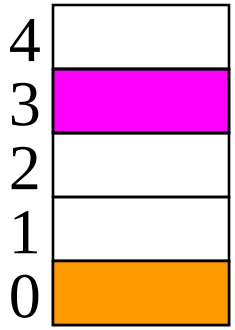








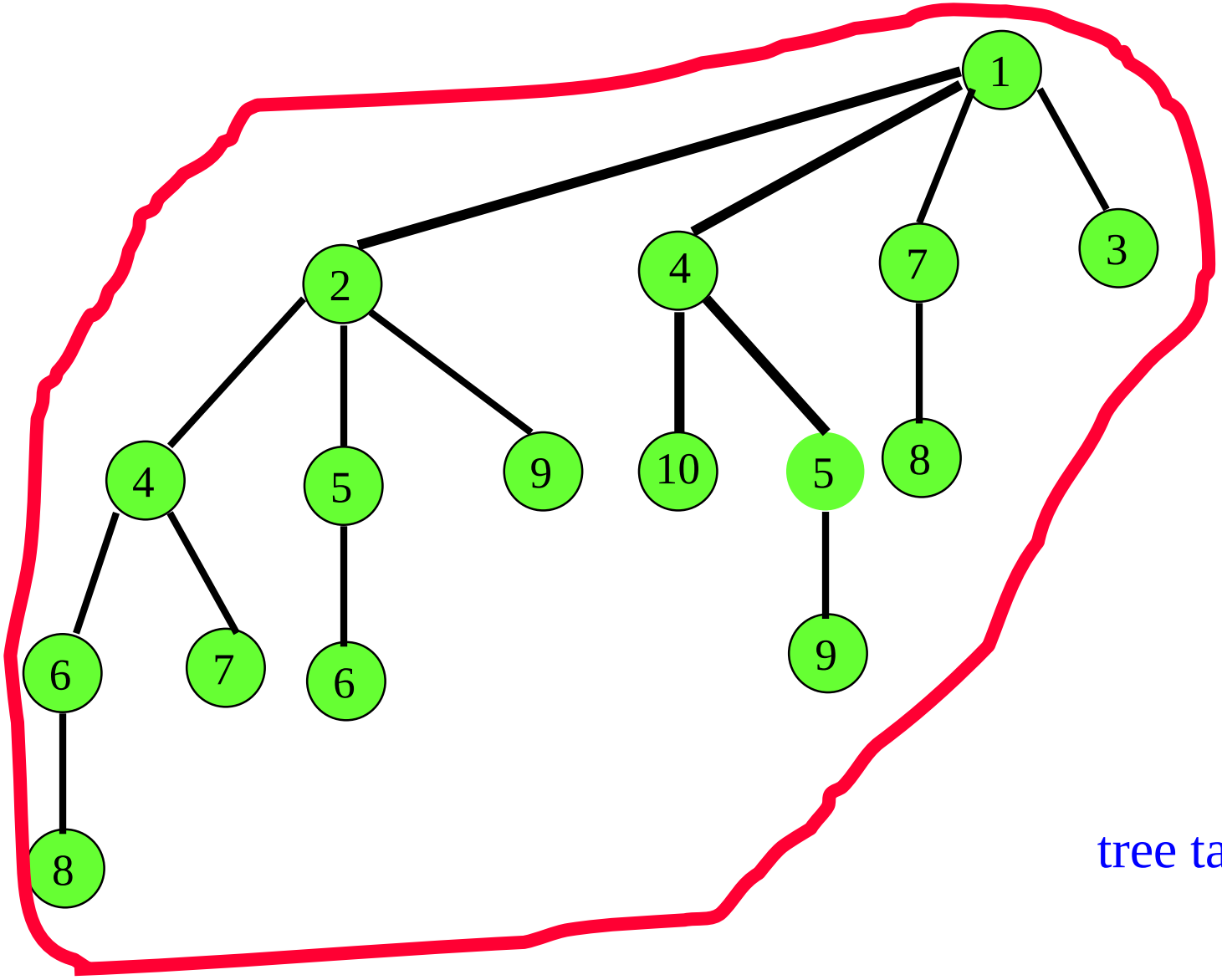
tree table

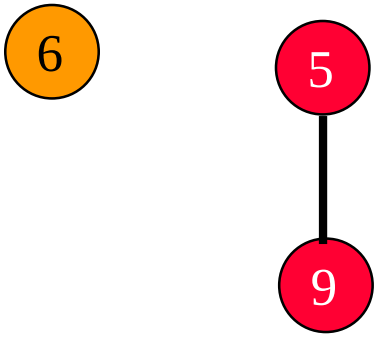


Combine 2 min trees of degree 3.

Make the one with larger root a subtree of other.

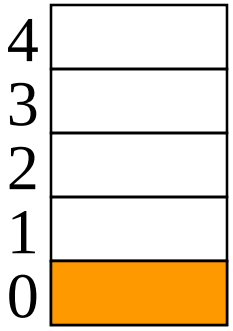
Pairwise Combine





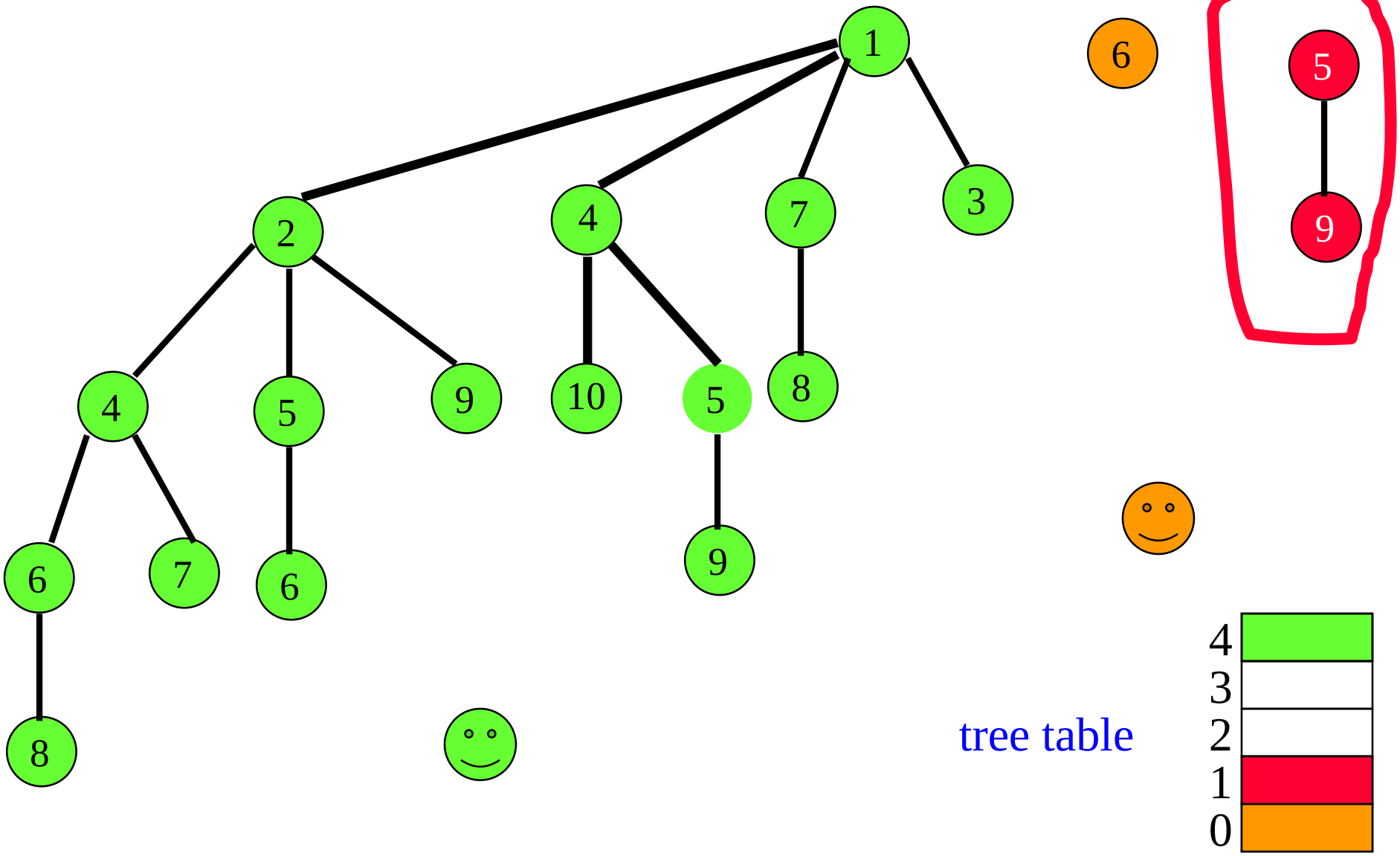


tree table

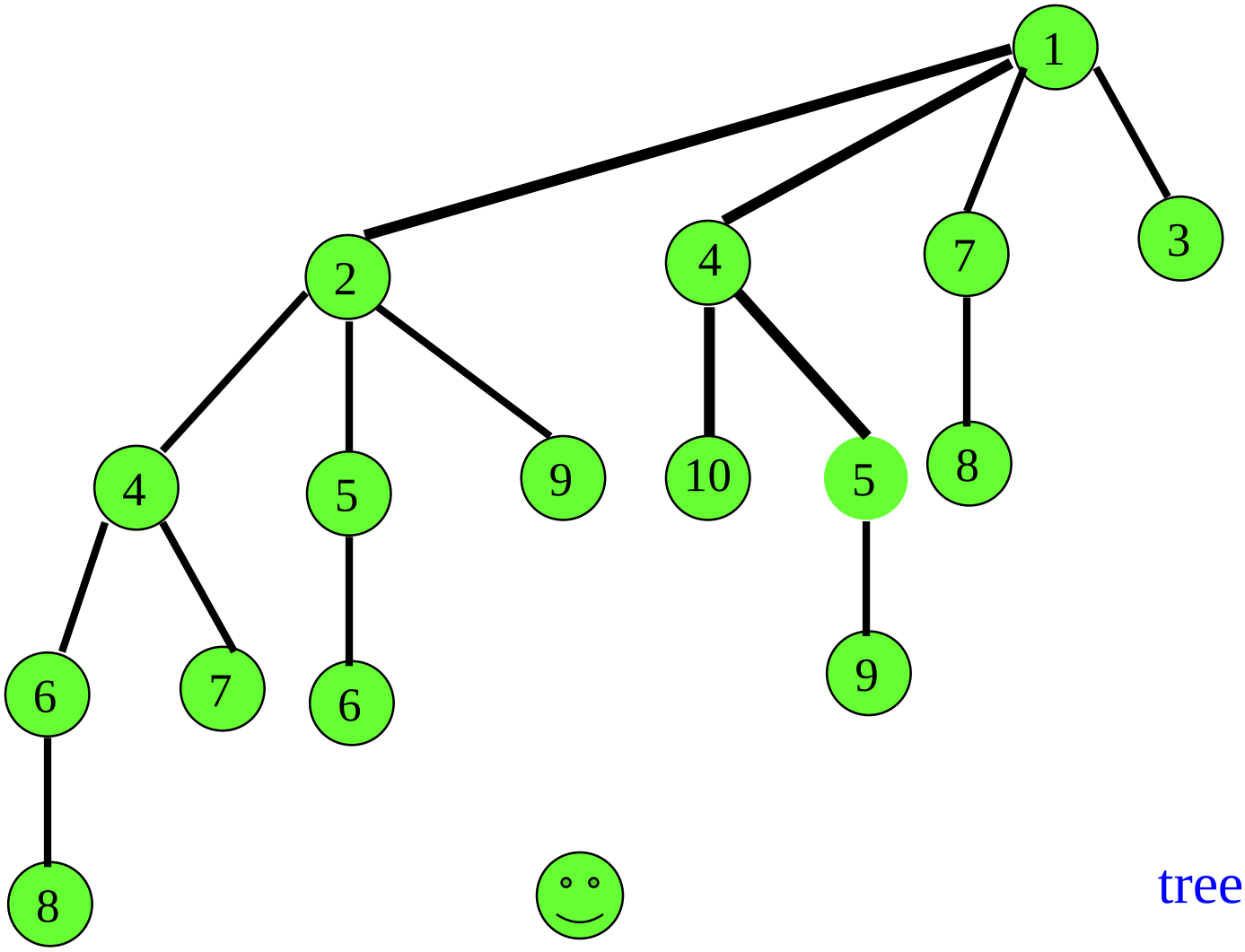


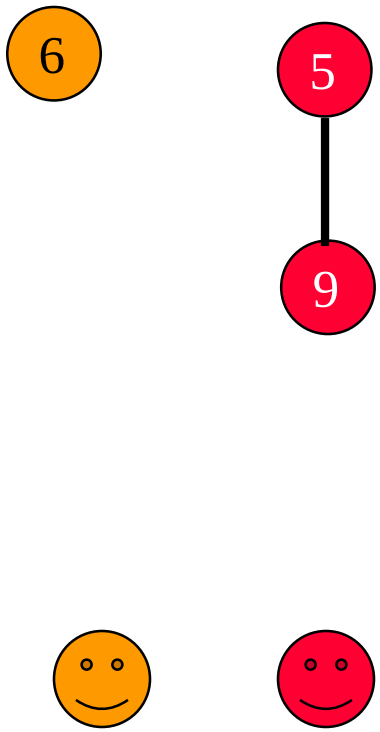
Update tree table.

Pairwise Combine



Pairwise Combine





tree table



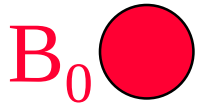
Create circular list of remaining trees.

Complexity Of Remove Min

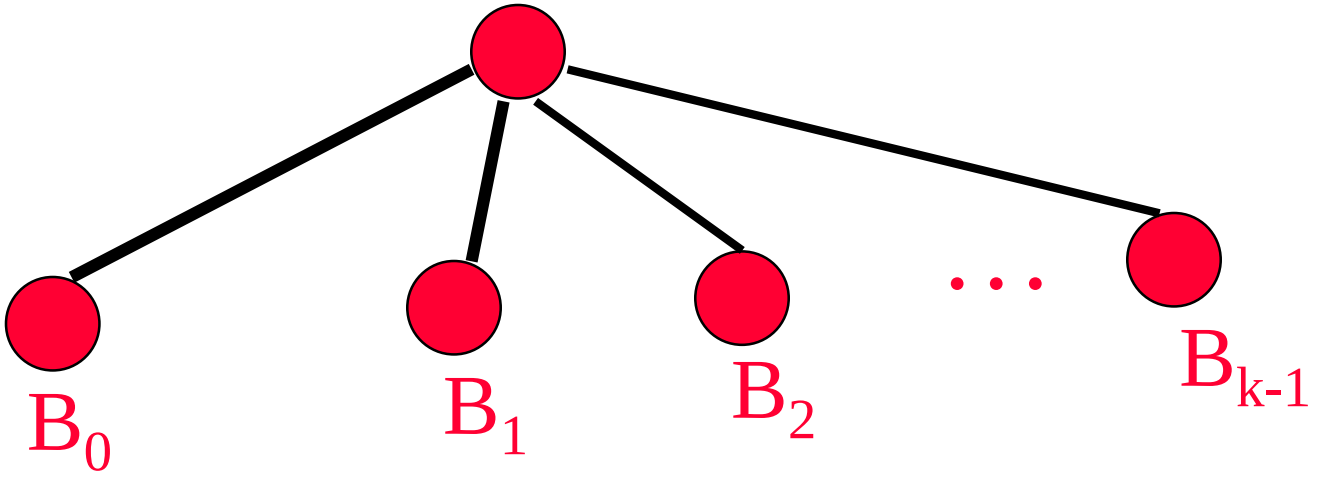
- Create and initialize tree table.
  - O(MaxDegree).
  - Done once only.
- Examine s min trees and pairwise combine.
  - O(s).
- Collect remaining trees from tree table, reset table entries to null, and set binomial heap pointer.
  - O(MaxDegree).
- Overall complexity of remove min.
  - O(MaxDegree + s).

Binomial Trees

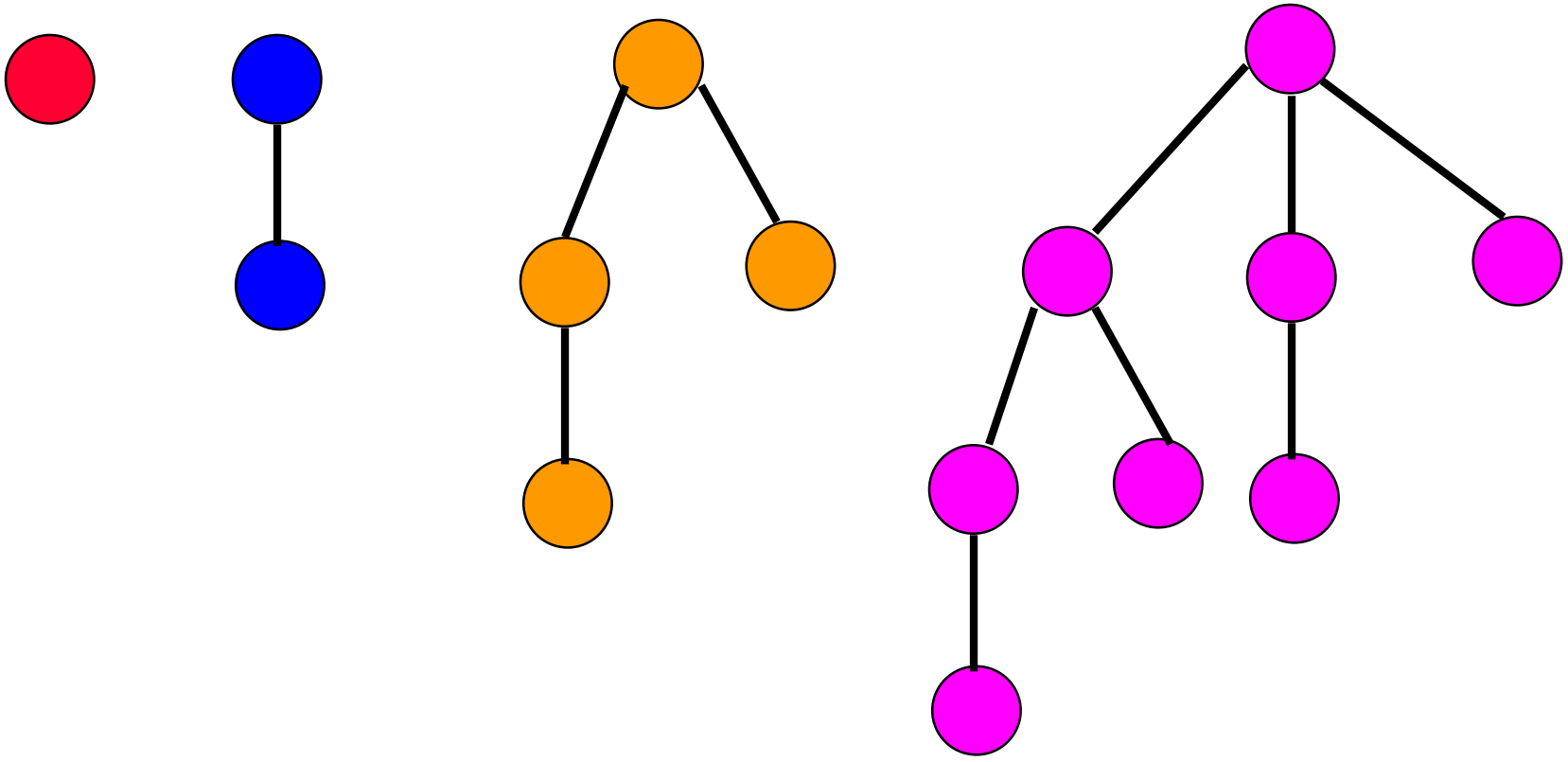
- B_k is degree k binomial tree.



- B_k , $k > 0$, is:

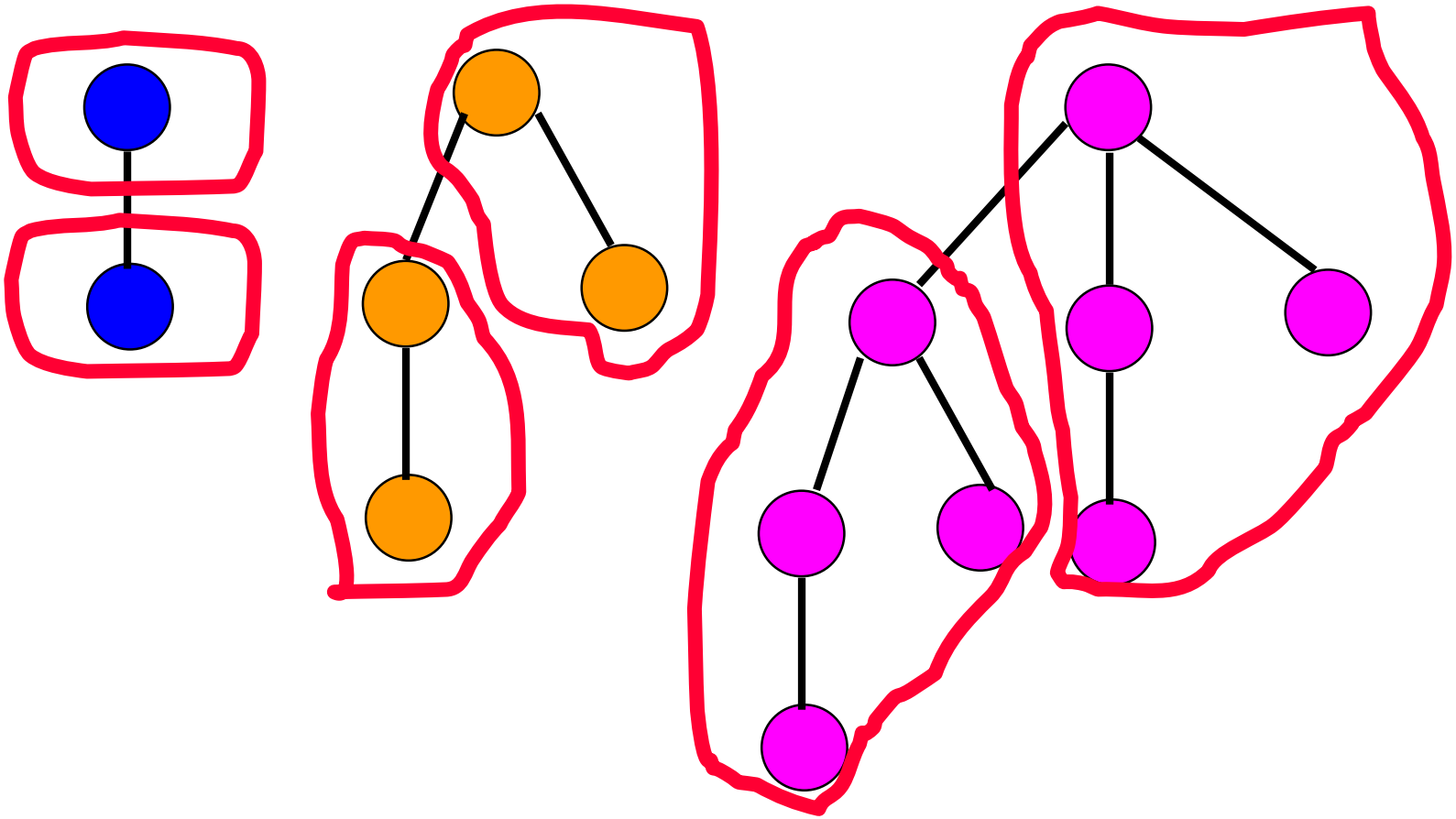


Examples



Number Of Nodes In B_k

- B_k , $k > 0$, is two B_{k-1}s.
- One of these is a subtree of the other.



N_k And MaxDegree

- $N_0 = 1$
- $N_k = 2N_{k-1}$
  $= 2^k$.
- If we start with zero elements and perform operations as described, then all trees in all binomial heaps are binomial trees.
- So, MaxDegree = O(log n).

Acknowledgements

- E. Horowitz, S. Sahni, and D. Mehta, Fundamentals of data structures in C++, Second Edition, Silicon Press, 2007.
- Lecture Slides of Prof Sartaj Sahni.